

OE-EC804A — Artificial Intelligence

Semester VIII • 3L:0T:0P • 3 Credits

Module 5: Knowledge Representation & Logical Agents

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1. Knowledge-Based Agents & Wumpus World

Definition of a Knowledge-Based Agent

A **Knowledge-Based Agent** maintains an internal **Knowledge Base (KB)** expressed in a formal representation language. It operates by maintaining sentences in its KB, inserting new percepts via `TELL`, and deriving actions via `ASK`.

Core Operations of KB Agent

- `TELL(KB, sentence)`: Adds new knowledge or percept to KB.
- `ASK(KB, query)`: Queries KB to infer whether query logically follows from KB.

1.1 The Wumpus World Environment

The **Wumpus World** is a benchmark domain for logical reasoning.

- 4×4 grid of rooms containing a Wumpus, Pits, and Gold.
- **Percepts**: [*Stench*, *Breeze*, *Glitter*, *Bump*, *Scream*].
- **Rules**: Rooms adjacent to Wumpus have *Stench*. Rooms adjacent to Pits have *Breeze*. Room containing Gold has *Glitter*.

Wumpus World Grid Environment (4×4)

Stench	Safe	Breeze	PIT
WUMPUS	GOLD	PIT	Breeze
Stench	Safe	Breeze	Safe
Start (1,1) Safe	Breeze	PIT	Breeze

2. Propositional Logic

Propositional logic represents facts using proposition symbols (P, Q, R).

Logical Entailment

Sentence α **entails** sentence β ($\alpha \models \beta$) if and only if in every model in which α is true, β is also true.

2.1 Logical Equivalences

Equivalence Law	Formula
Implication Elimination	$\alpha \implies \beta \equiv \neg\alpha \vee \beta$
Biconditional Elimination	$\alpha \iff \beta \equiv (\alpha \implies \beta) \wedge (\beta \implies \alpha)$
De Morgan's Laws	$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta \quad ; \quad \neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$
Distributivity	$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$

3. Inference & Resolution in Propositional Logic

3.1 Resolution Rule

Unit Resolution rule states that:

$$\frac{l_1 \vee l_2 \vee \dots \vee l_k, \quad \neg l_i}{l_1 \vee \dots \vee l_{i-1} \vee l_{i+1} \vee \dots \vee l_k} \quad (1)$$

Resolution Refutation proves $KB \models \alpha$ by showing that $KB \wedge \neg\alpha$ is **unsatisfiable**.

3.2 Conjunctive Normal Form (CNF) Conversion Steps

1. Eliminate $\iff$ replacing $\alpha \iff \beta$ with $(\alpha \implies \beta) \wedge (\beta \implies \alpha)$.
2. Eliminate $\implies$ replacing $\alpha \implies \beta$ with $\neg\alpha \vee \beta$.
3. Move $\neg$ inwards using De Morgan's laws ($\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$).
4. Distribute $\vee$ over $\wedge$ ($\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$).

Quick Revision — Module 5

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Key Concepts and Metrics at a Glance

Logical Entailment	$KB \models \alpha \iff M(KB) \subseteq M(\alpha) \quad (\alpha \text{ true in all KB models})$
Proof by Contradiction	$KB \models \alpha \iff (KB \wedge \neg\alpha) \text{ is unsatisfiable}$
Implication Elimination	$\alpha \implies \beta \equiv \neg\alpha \vee \beta$
Resolution Rule	$\frac{\alpha \vee \beta, \neg\beta \vee \gamma}{\alpha \vee \gamma} \quad (\text{Derives empty clause on contradiction})$
CNF Structure	Conjunction of disjunctions of literals (AND of OR clauses)

One-Line Definitions

- **Knowledge Base (KB):** A set of sentences expressed in a formal representation language.
- **Model:** A mathematical world assigning truth values (True/False) to every proposition.
- **CNF:** A conjunction of disjunctions of literals (AND of ORs).
- **Sound Inference:** Derives only entailed sentences (no false conclusions).
- **Complete Inference:** Can derive any sentence that is entailed.

Frequently Asked / Viva Questions

Q1. Define a Knowledge-Based Agent.

An agent that reasons over a formal Knowledge Base (KB) using TELL and ASK operations to select actions.

Q2. What is Entailment ($\models$)?

$KB \models \alpha$ means α is true in all models where KB is true.

Q3. What is the difference between Syntax and Semantics?

Syntax defines allowable sentence structures. Semantics defines sentence meaning and truth values in models.

Q4. State De Morgan's Laws.

$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ and $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$.

Q5. What is Conjunctive Normal Form (CNF)?

A logical sentence format structured as an AND of clauses, where each clause is an OR of literals.

Q6. How does Resolution Refutation work?

Frequently Asked / Viva Questions (contd.)

To prove $KB \models \alpha$, add $\neg\alpha$ to KB and apply resolution rules until an empty clause (contradiction) is derived.

Q7. Differentiate Soundness and Completeness.

Soundness: Everything derived is true ($KB \vdash \alpha \implies KB \models \alpha$). Completeness: Every true fact can be derived ($KB \models \alpha \implies KB \vdash \alpha$).

Q8. What is Forward Chaining?

An inference algorithm for Horn clauses that starts from known facts and applies rules to infer new facts until query is proved.

Q9. What is Backward Chaining?

An inference algorithm that starts from query goal and works backward through rules to find supporting facts.

Q10. What is a Horn Clause?

A disjunction of literals containing at most one positive literal.